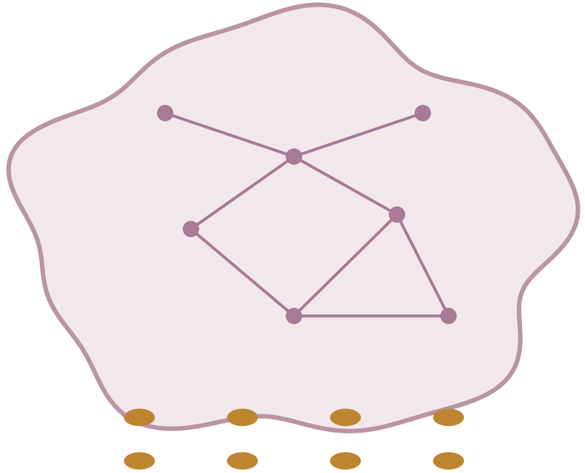
Can electrical activity help estimate culture age?

Original research aim: estimate age and investigate remaining functional life.



Living neural culture

MEA event recordings

Describe the activity

Counts and channel activity
Timing and bursts
Environmental measurements

Evaluate on held-out groups

Age

Requires a defined recording-age label

Functional lifespan

Requires follow-up and endpoint labels

Conceptual schematic. Tissue shape, electrode contacts and connections are illustrative. Results and available labels determine the supported claims.

Four data tracks answer different questions

Recording rows, clock windows and cultures represent different units of evidence.

Data track	Material / format	Available data	Analysis role
FinalSpark	Aggregate human cultures HDF5 event exports	FS369: 94.15 million events FS437: 2.64 million events	Exploratory age and recorded-endpoint models
DANDI 002020	Dissociated rat cultures NWB detected-spike tables	129 sessions 7 cultures	Descriptive activity across recording sessions
Zenodo 22755004	Human forebrain organoids MAT individual recordings	45 recordings 16 experiment prefixes	Descriptive activity with file-level quality checks
KCL culture time	Organoid-derived 2D cultures Published well-average CSV	920 recordings / 216 wells 6 lines / 15 line-batches	Days after dissociation Grouped age benchmark

The added age benchmark uses explicit DPD labels. The external descriptive tracks do not validate lifespan.

Sources: user-derived FinalSpark, DANDI 002020 and Zenodo 22755004 tables; KCL dataset DOI 10.18742/30394168. Event totals are detector-specific.

Two modeling paths share a reproducible workflow

FinalSpark features come from event exports. The culture-age extension uses published five-minute well summaries.

FinalSpark

HDF5 event tables

Read-only, 100,000 rows
per input chunk

Hourly features

Partition by clock hour
Sort within each channel

Training pipeline

Activity / environment
Median, Ridge, forest

Transfer evaluation

Train FS369
Evaluate FS437

Culture-age benchmark

Published source CSV

Join 920 record IDs
to verified metadata

Six activity features

Keep measured summaries
Exclude age and identifiers

Training pipeline

Fold-specific preprocessing
Fixed model settings

Grouped evaluation

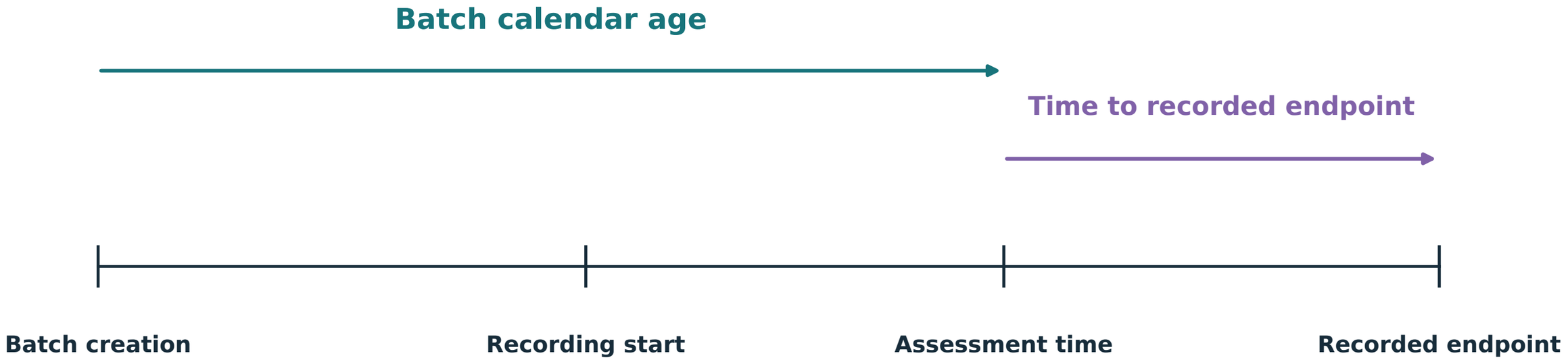
Hold out plates / batches
Then donor cell lines

Each run saves source audits, split memberships, predictions, metrics and software provenance.

Methods: BDAPP hourly extractor and models; age extension analysis plan. Imputation and scaling fit only on training data.

The FinalSpark targets follow the recorded experiment clock

Calendar batch age and time to the recorded endpoint share the same assessment timestamp.



Age = assessment time − reported batch creation

Remaining recorded time = endpoint − assessment time

Source: FinalSpark metadata. Endpoint causes: "Almost no activity" and "No activity". This schematic does not show independently verified individual deaths.

Activity features summarize several aspects of the recording

These are measured event summaries. Their predictive value comes from held-out evaluation.

01

Counts

Event count
Active observed channels

02

Channel distribution

Entropy of channel shares
Largest channel event share

03

Amplitude

Absolute median
Absolute interquartile range

04

Timing

Within-channel intervals
CV, CV2 and short intervals

05

Channel bursts

At least 5 linked events
Intervals at most 100 ms

06

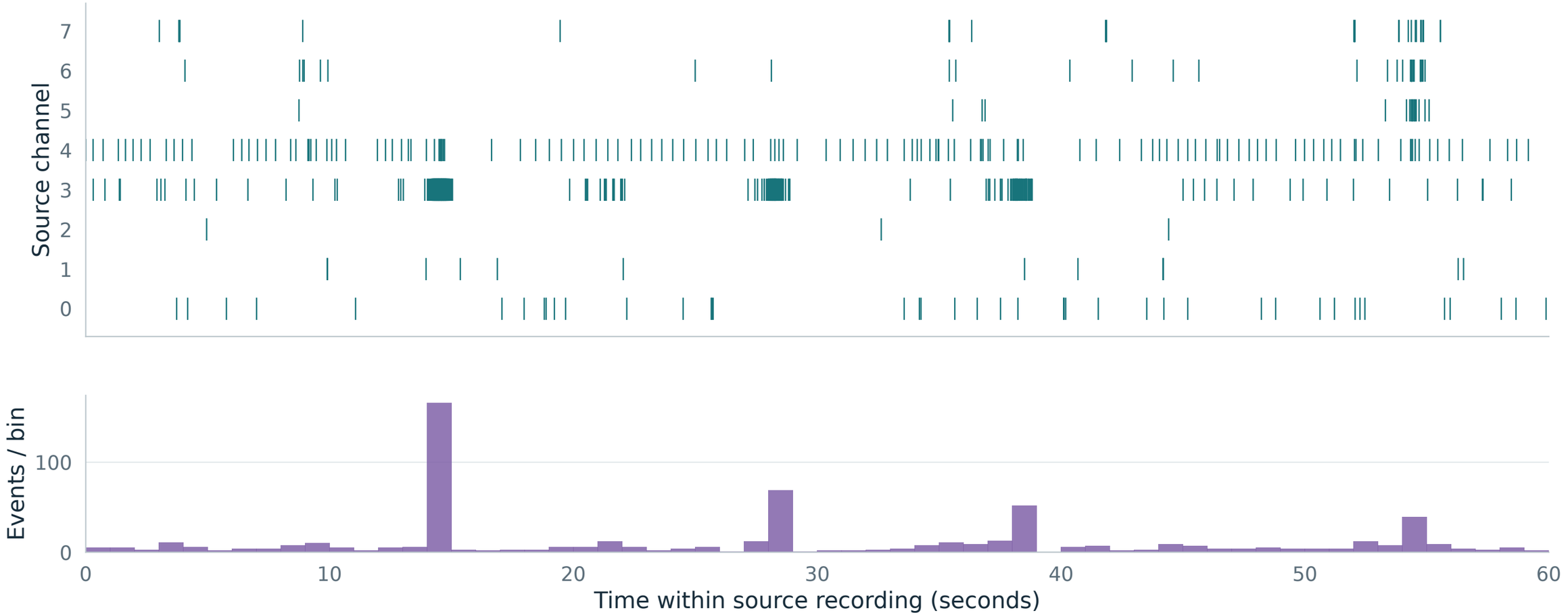
Environment

Temperature, CO_2 and O_2
Exported stimulation logs

FinalSpark hourly features: counts, channel distribution, amplitude, timing and fixed-ISI bursts. Environmental columns remain source-reported values.

A real event raster shows the input to activity analysis

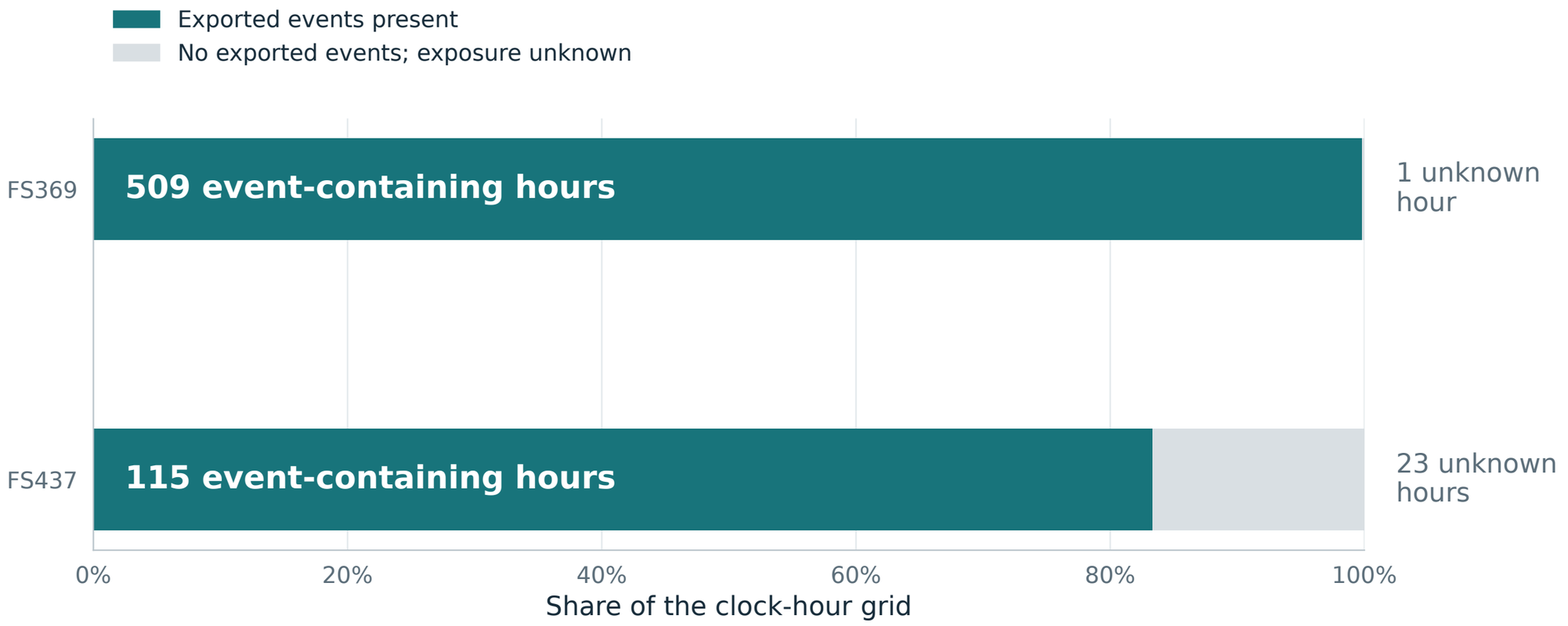
Zenodo fs363-org0, first 60 seconds. Each tick is one detected event on a source channel.



Source: Zenodo 22755004, fs363-org0.mat. Actual spike times, no waveform data. The event-count panel uses one-second bins.

Event-free export windows have unknown observation status

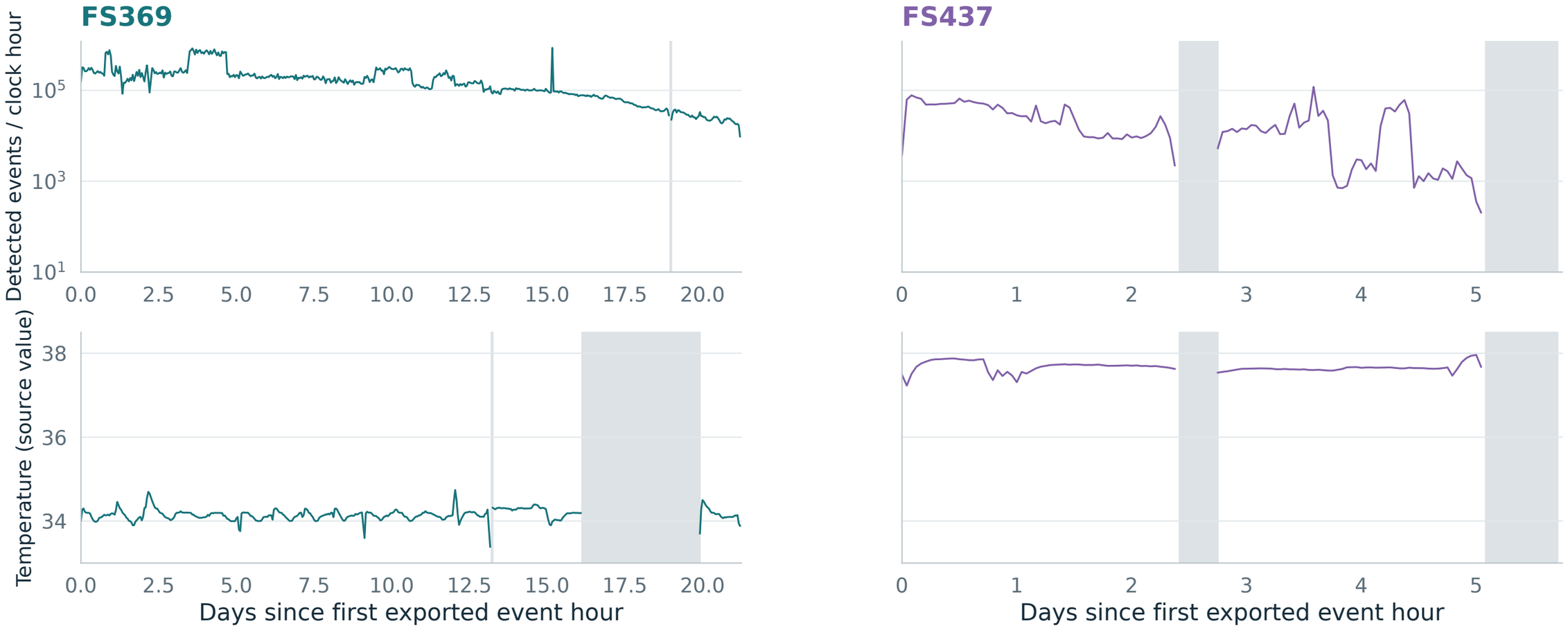
The hourly grids contain 510 clock hours for FS369 and 138 for FS437.



Source: FinalSpark hourly tables. Exported-event presence does not establish continuous recording exposure. Missing event records do not establish physiological silence.

Activity and source-reported temperature differ between experiments

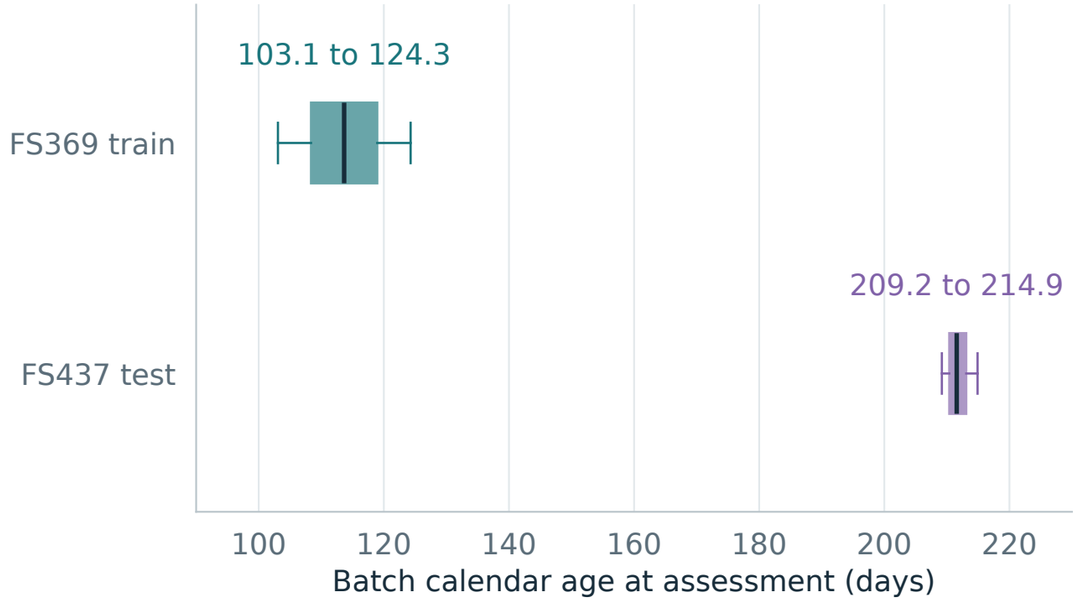
Hourly exports retain missing values. Counts are not normalized by verified recording exposure.

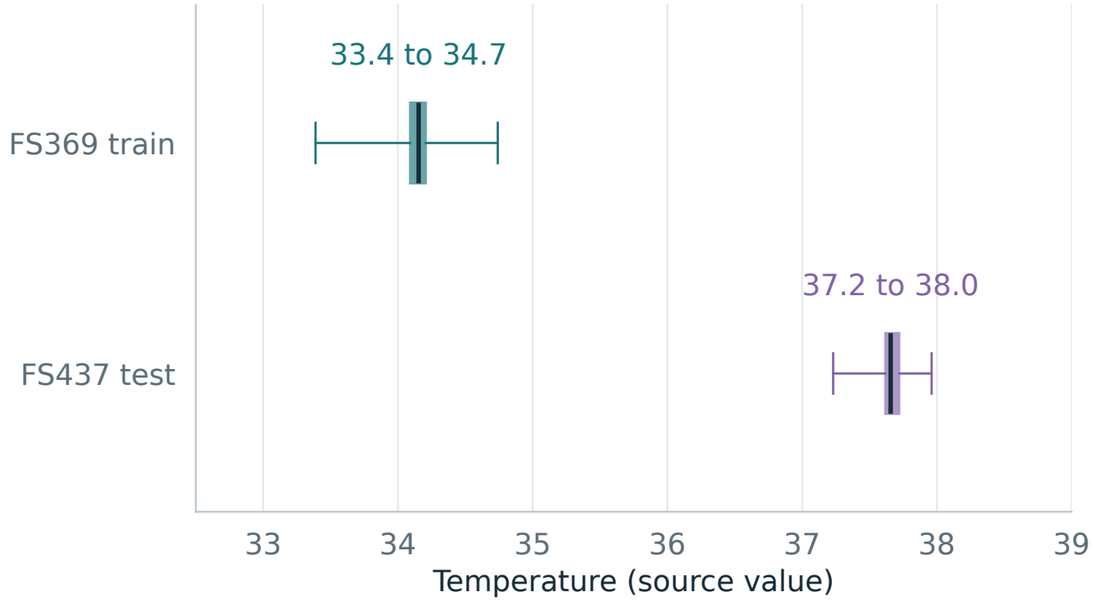


Source: user-derived FinalSpark hourly features. Gray shading marks absent exported values in the plotted quantity. Environmental differences do not establish causal effects.

FS437 falls outside the training age range

The transfer test combines age extrapolation with a different acquisition and environmental context.

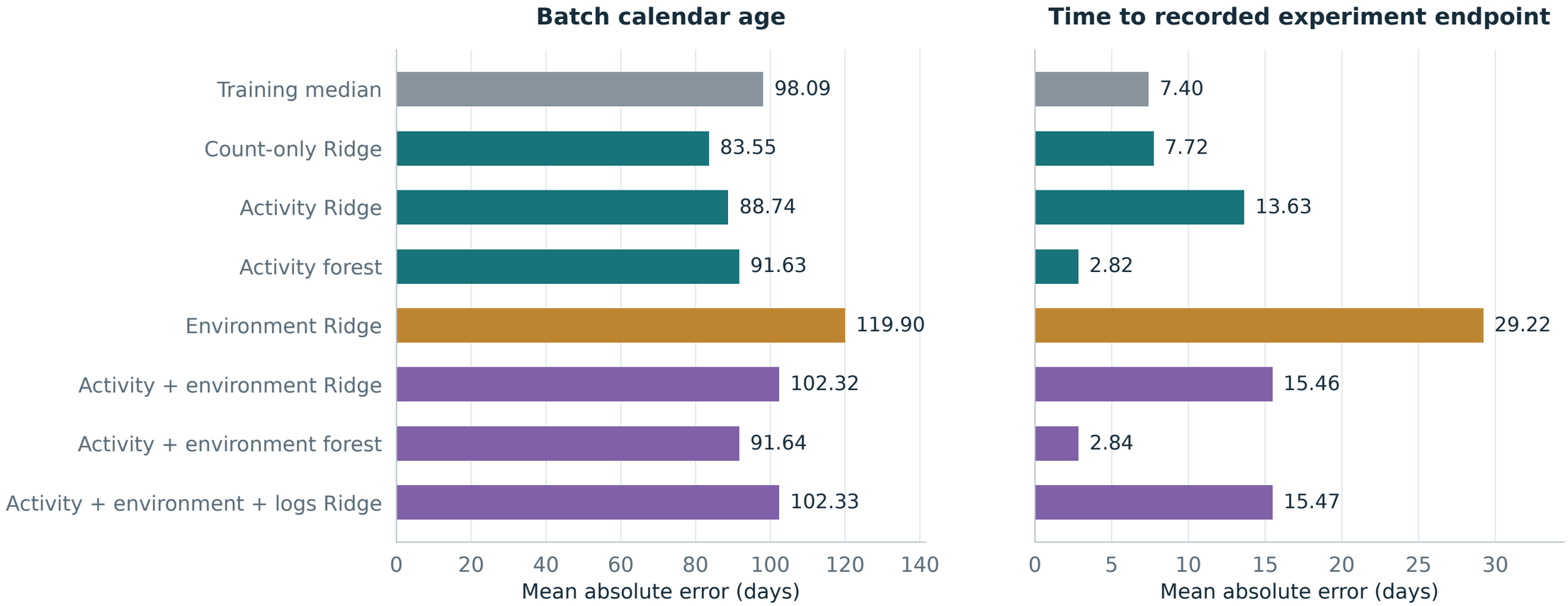




Source: model-eligible FinalSpark hours, assessed at window end. Boxplots show medians, quartiles and full observed ranges. Temperature retains source-reported units.

FinalSpark transfer remains poor across the primary models

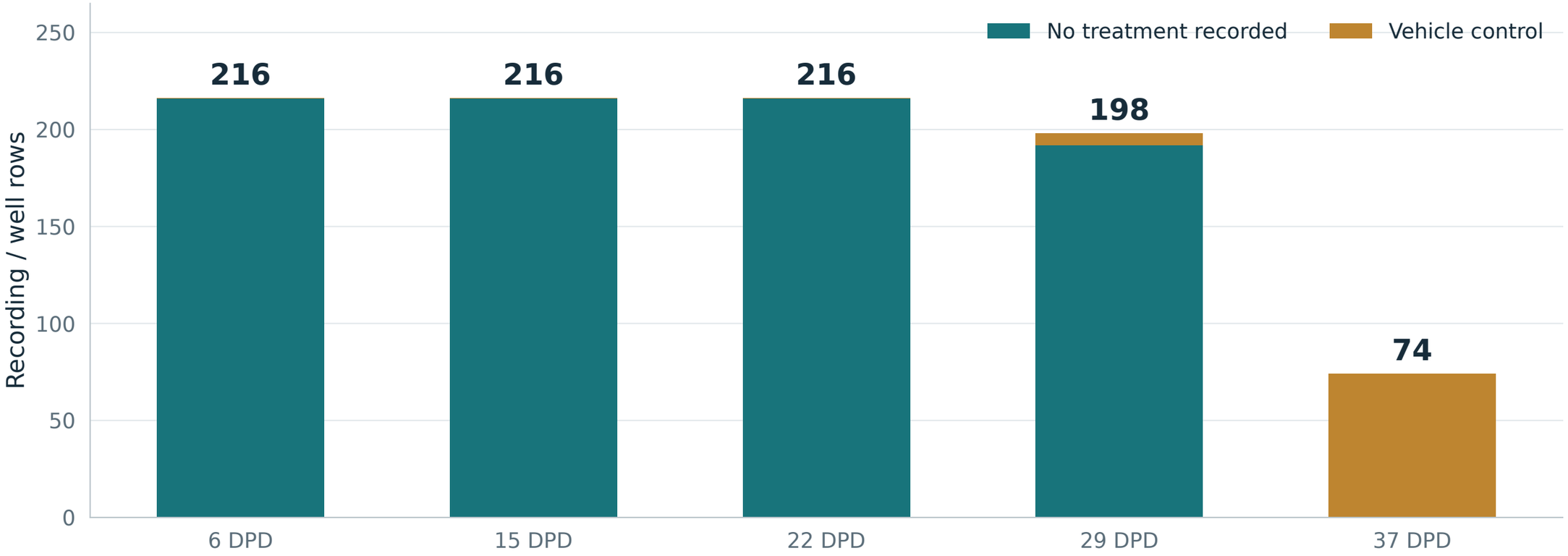
Train on 509 FS369 hours. Evaluate on 115 FS437 hours. Every primary transfer R^2 is negative.



Source: final successful Windows run reported by the user, 27 October 2026. Two different experiment-level targets; neither measures individual biological lifespan.

The age benchmark has explicit labels at five culture times

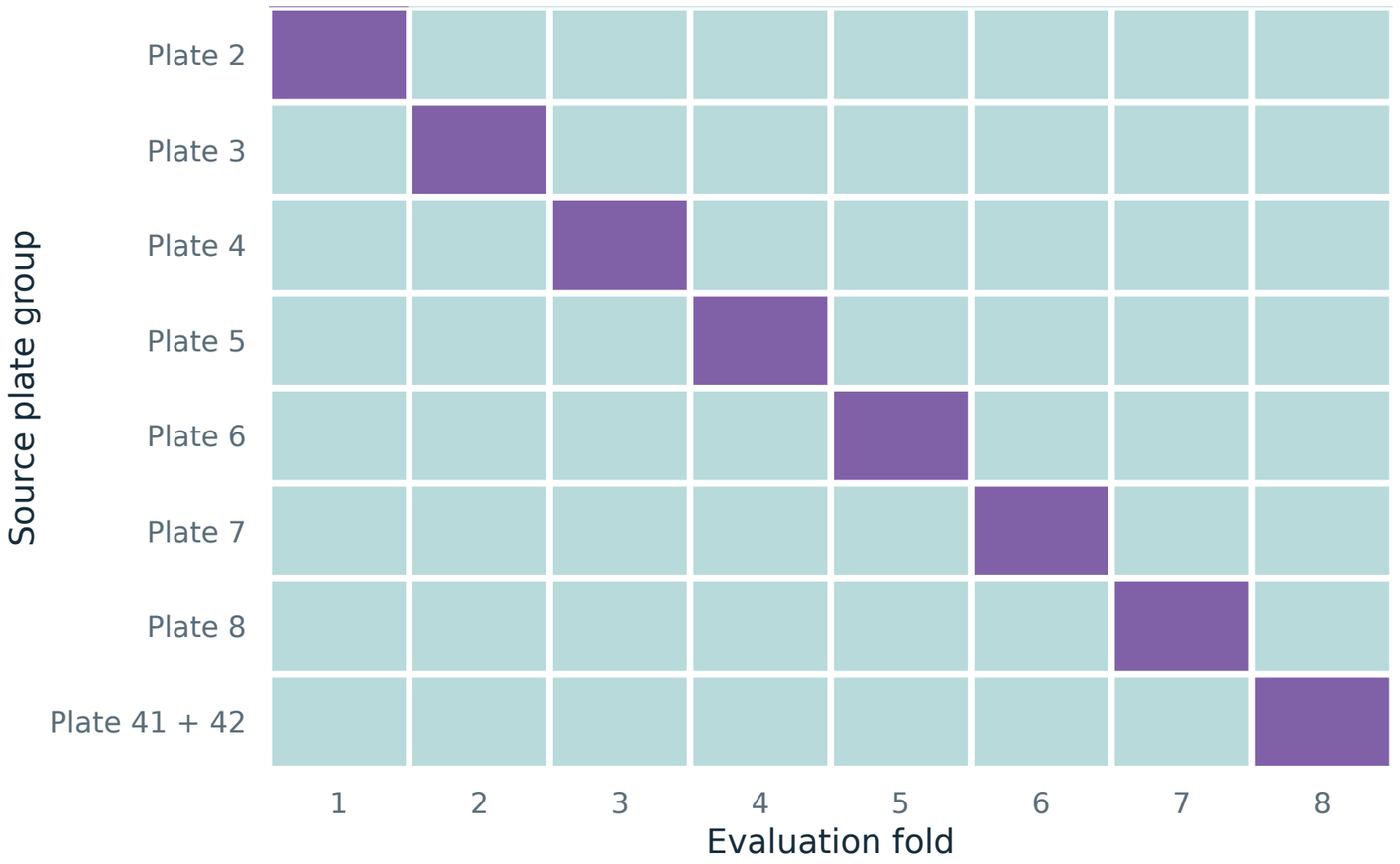
920 five-minute recordings from 216 repeated wells, six donor lines and 15 line-batch combinations.



Source: KCL DOI 10.18742/30394168, development time course with MEA-NAP features and joined metadata. DPD = days post dissociation.

Whole plate and batch groups stay outside training

Eight connected groups prevent shared wells, plates and differentiation batches across each split.



Held-out group

Training groups

216 repeated wells

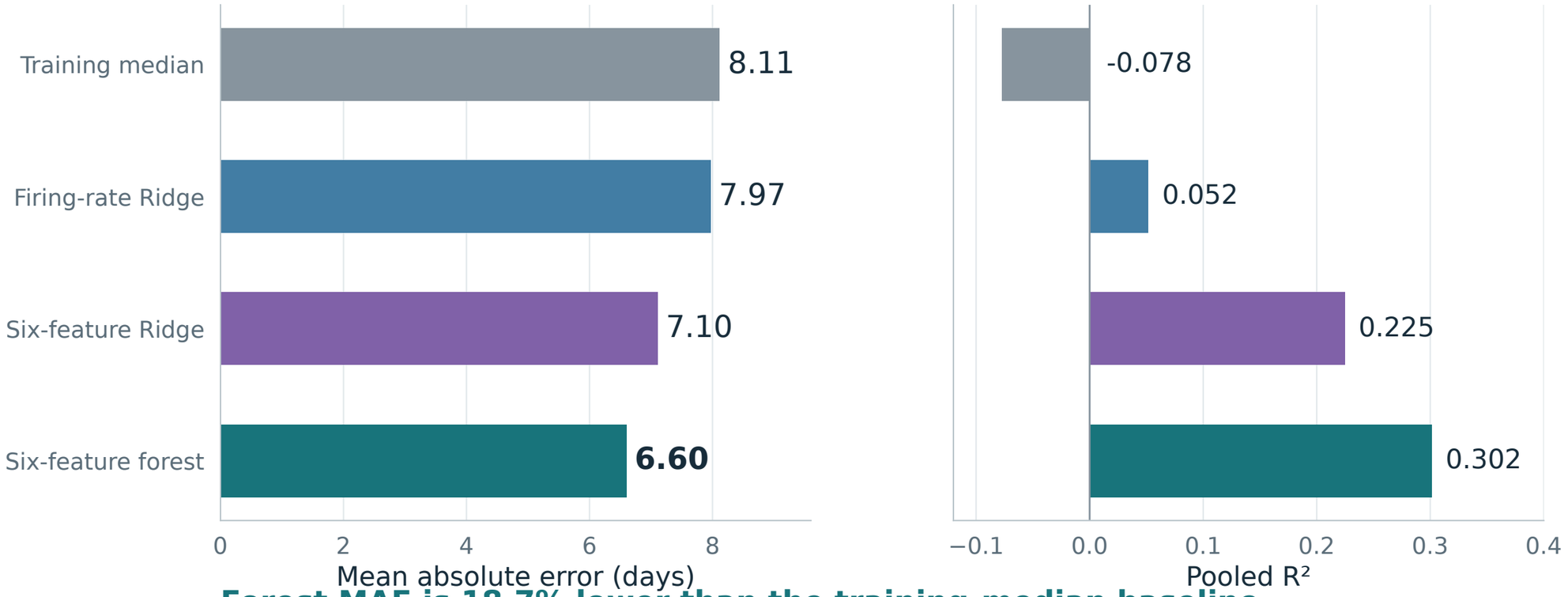
15 line-batches

9 physical plates

Source: age benchmark fold membership. Plates 41 and 42 share a line-batch and remain one group. The secondary donor-line test also excludes linked plates from training.

Activity features provide a modest culture-time prediction signal

Primary evaluation: all 920 recording rows, with whole plate/batch groups held out.

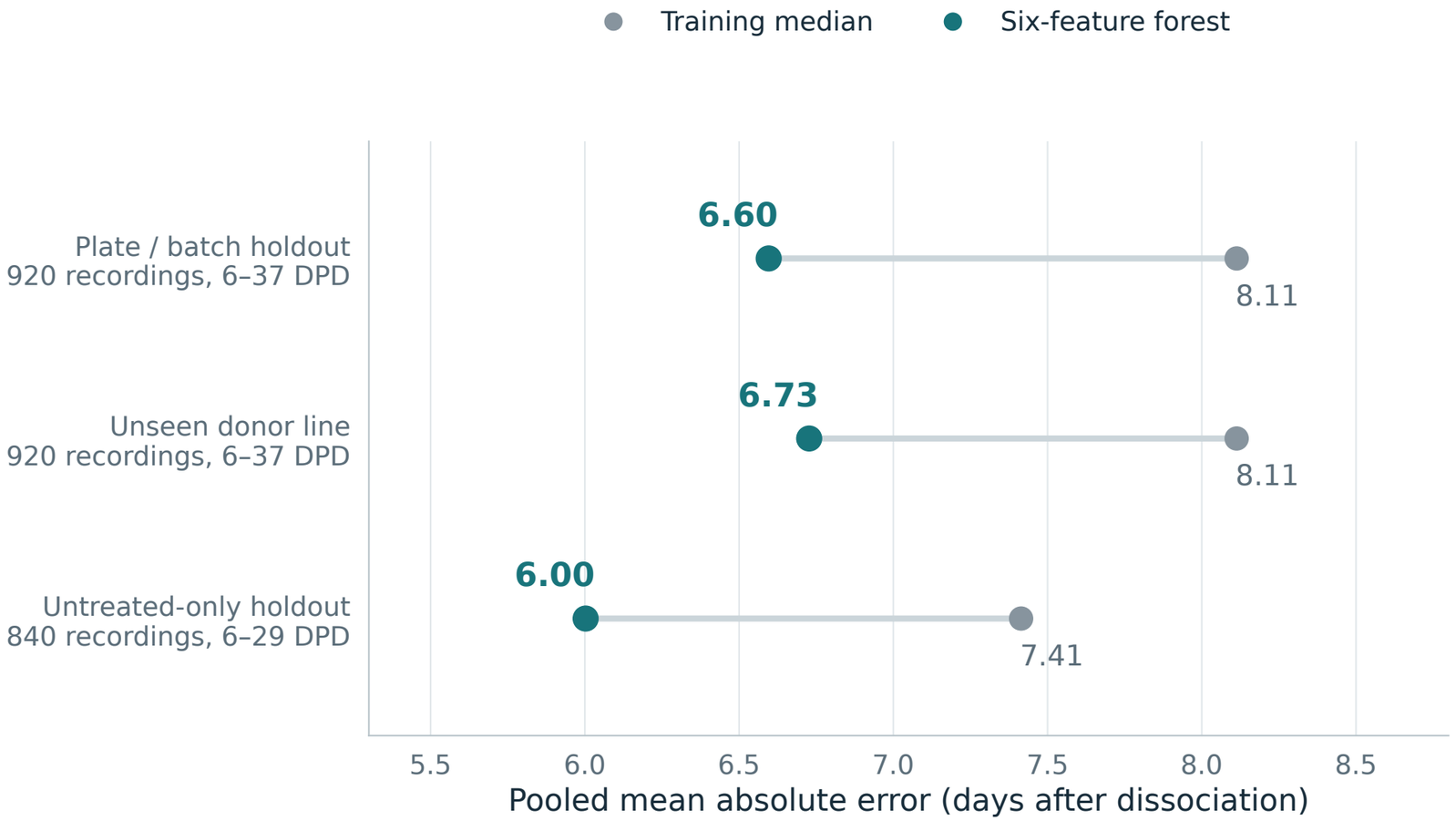


Forest MAE is 18.7% lower than the training-median baseline.

Source: verified culture-age benchmark. Target = days after dissociation from five-minute well summaries. Pooled cross-validation scores; no prospective external validation.

The age signal persists under donor-line exclusion

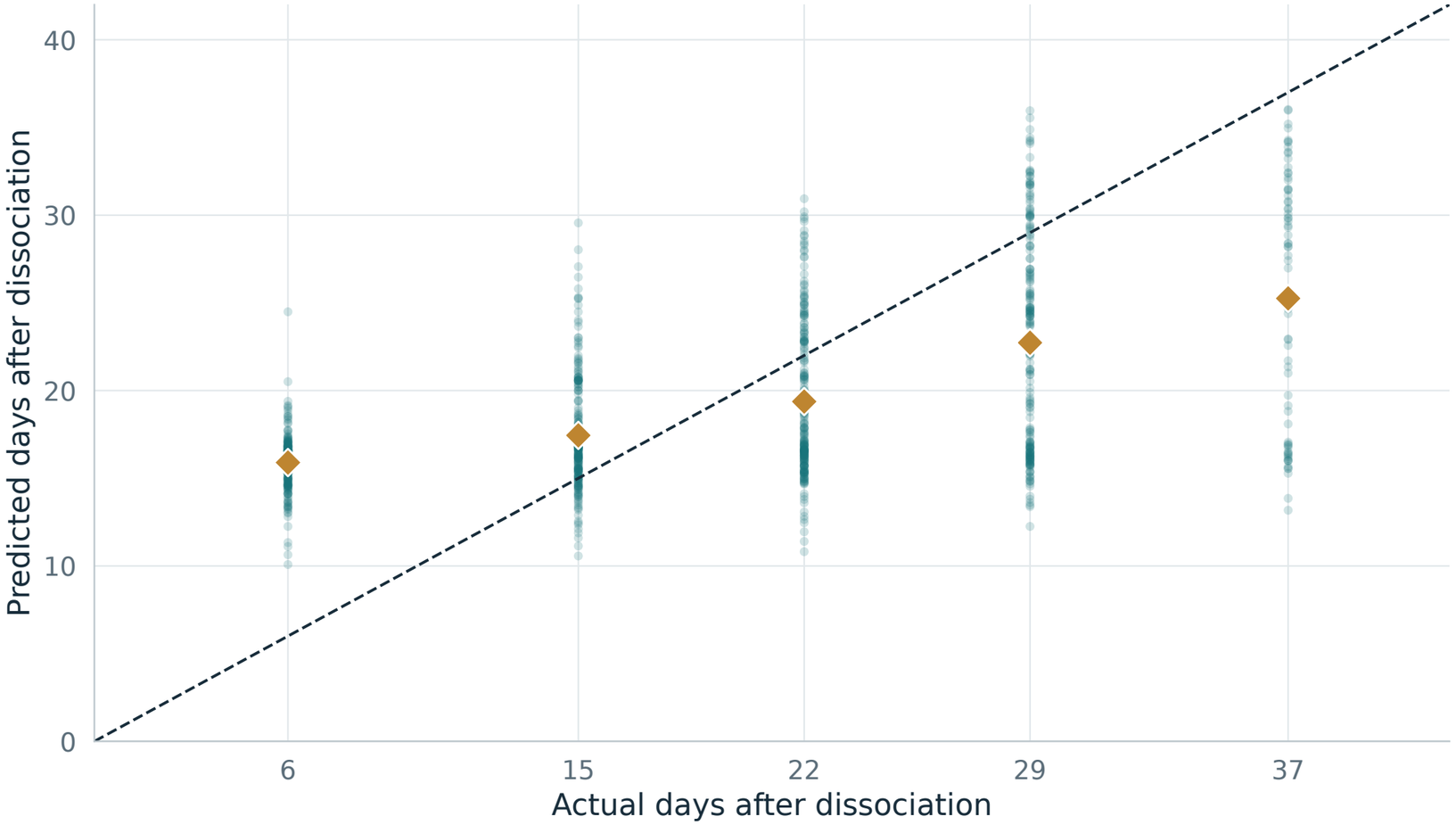
The untreated-only check uses a narrower population and age range.



Source: all three predeclared culture-age protocols. Dots show pooled MAE, not confidence intervals. Untreated-only errors are not directly comparable to the full-age test.

Held-out predictions recover some culture-time structure

Each point is a recording row. Repeated wells contribute observations at several sampled ages.



6.60 days

Mean absolute error

R^2 = 0.302

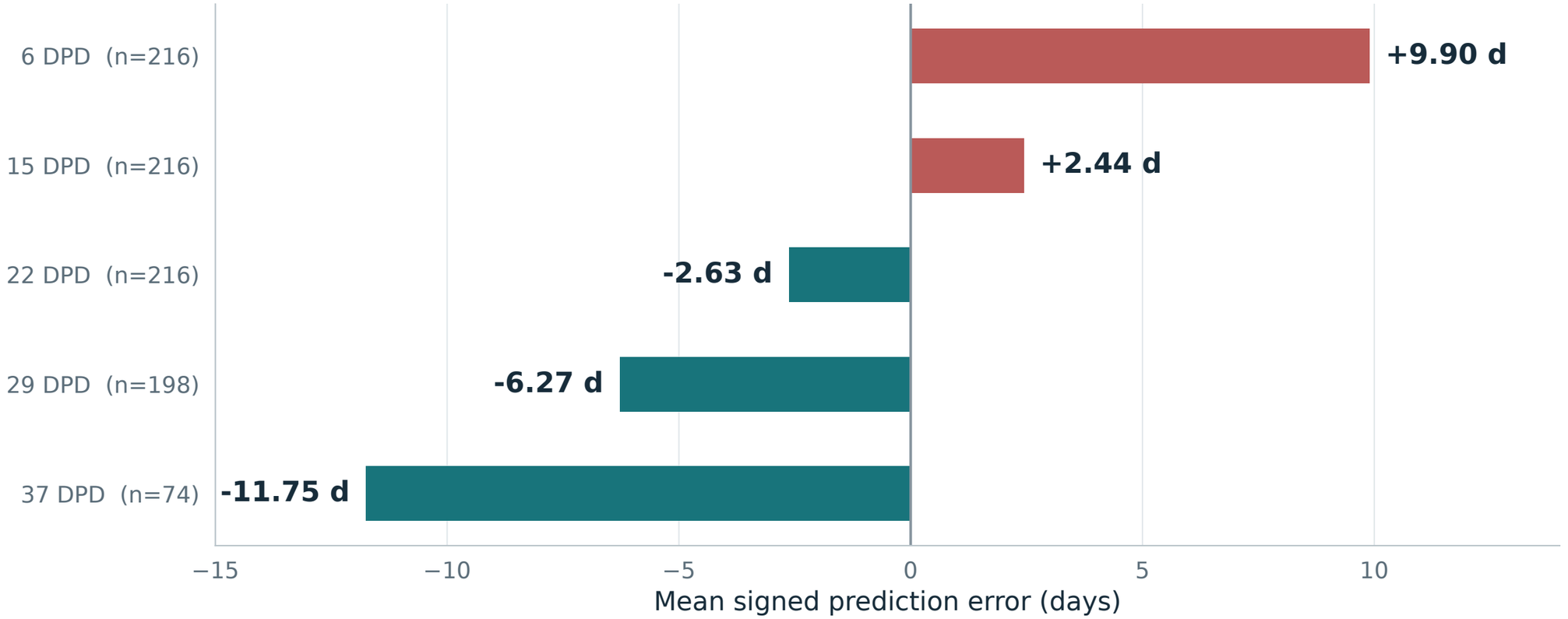
Whole plate/batch groups held out

Accuracy remains limited at the youngest and oldest sampled ages.

Source: primary grouped forest, 920 out-of-fold predictions. Diamonds mark mean predictions at each age. The dashed line marks perfect prediction.

Early ages are overestimated and late ages underestimated

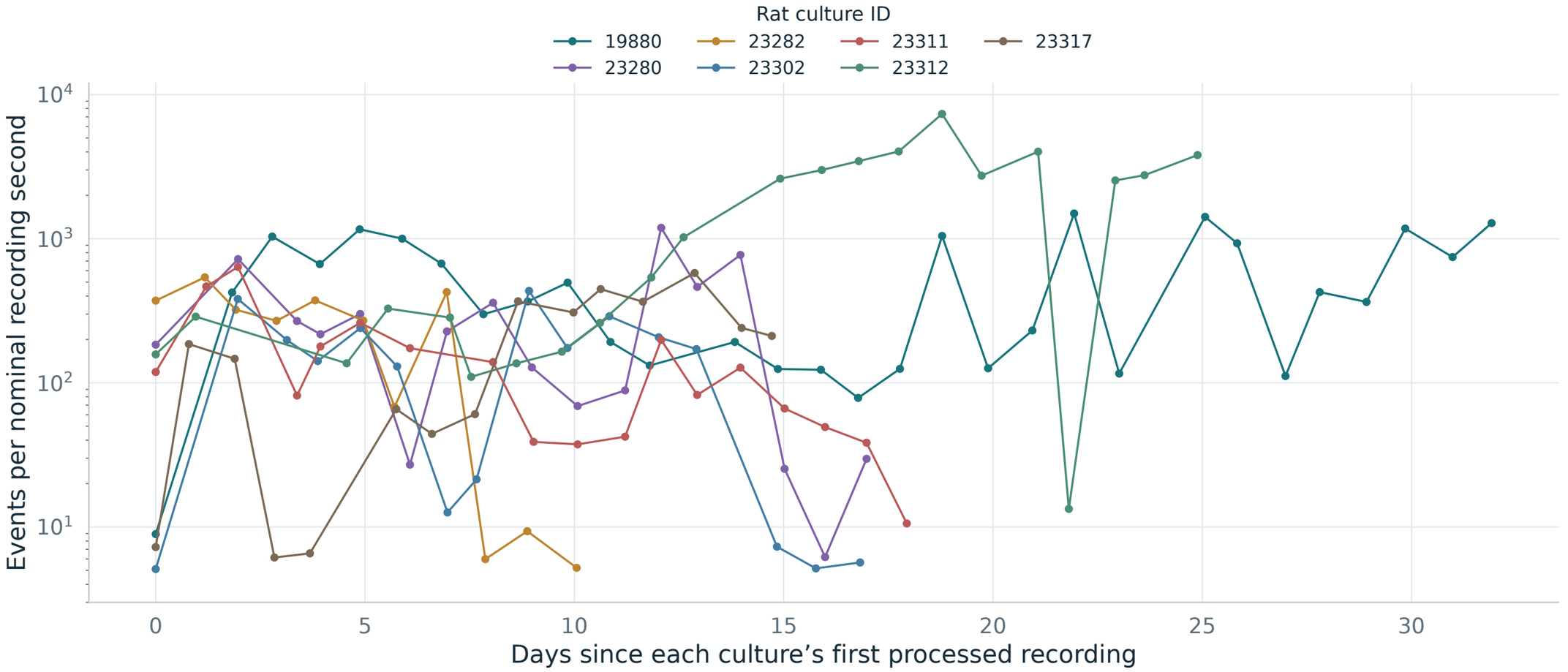
Mean signed error = predicted age minus actual age, using the primary grouped forest.



Source: all primary forest predictions, grouped by actual DPD. This plot shows average bias, not confidence intervals. Pooled MAE alone does not describe calibration.

DANDI adds longitudinal descriptive evidence from rat cultures

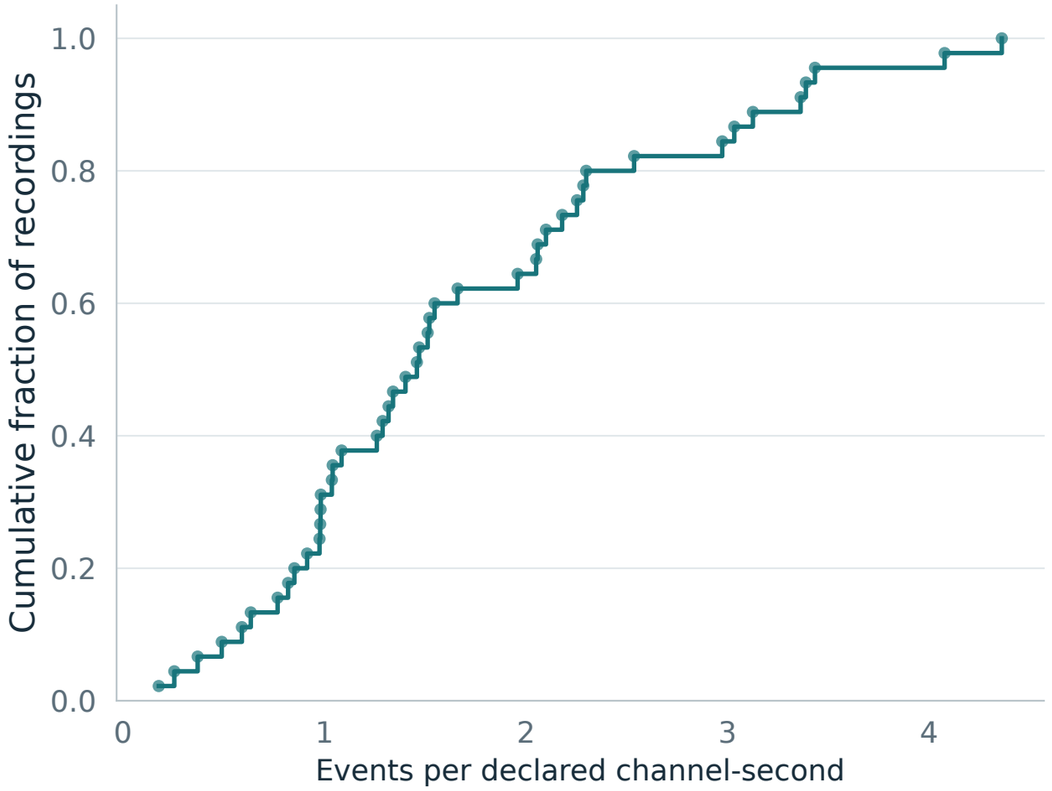
129 recording sessions from seven cultures. Each point marks an available session.

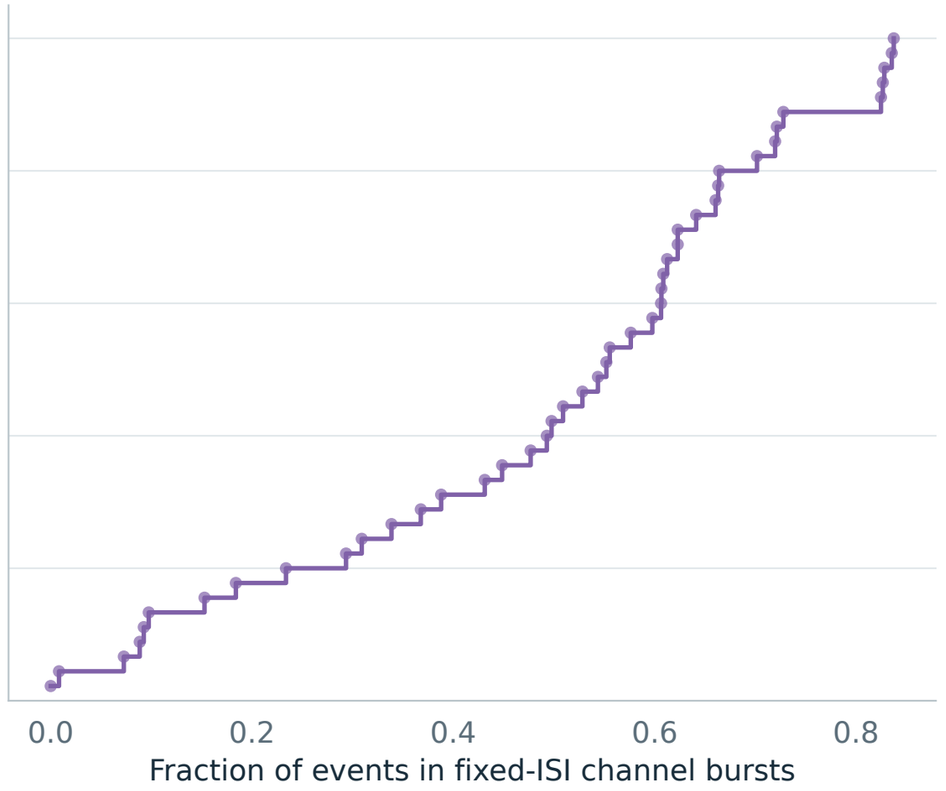


Source: DANDI 002020, user-processed NWB detected-spike tables. Rates use converter-reported nominal duration. Elapsed recording time is not culture age.

Zenodo summarizes activity in 45 human organoid recordings

Sixteen experiment prefixes may contain related recordings. The individual age/death labels remain unavailable.





Source: Zenodo 22755004. Rates use file-declared durations and channel counts. fs417-org0 retains 33,096 event times while its mismatched amplitudes remain missing.

The evidence supports a modest culture-time benchmark

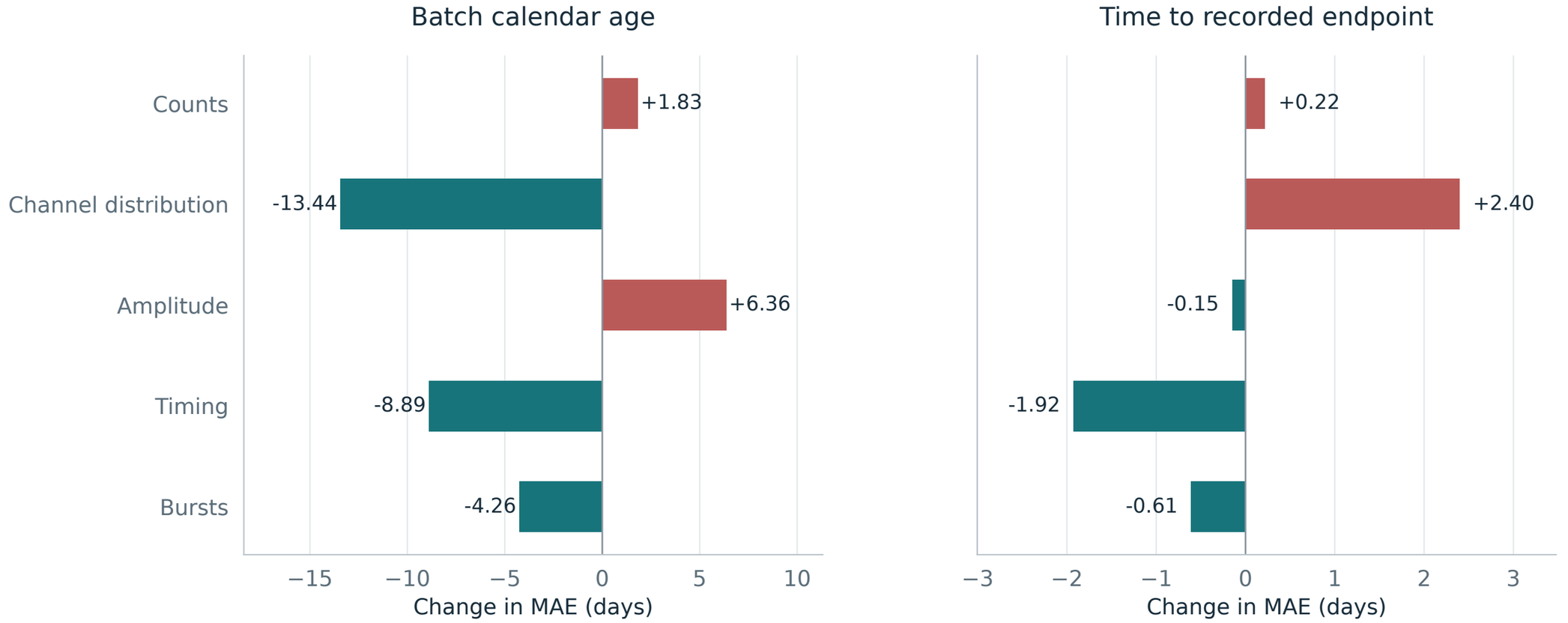
The project combines reproducible data engineering with explicit limits on prediction claims.

Data pipeline	Large event exports processed and audited
FinalSpark transfer	Primary models have negative R^2 on FS437
Culture-time benchmark	Grouped forest MAE 6.60 DPD, pooled R^2 0.302
Remaining functional life	Needs suitable individual follow-up and endpoint labels

Source: verified BDAPP extraction/model checks and separate KCL culture-time benchmark. Neither study validates individual remaining functional lifespan.

Feature ablations do not resolve the FinalSpark transfer problem

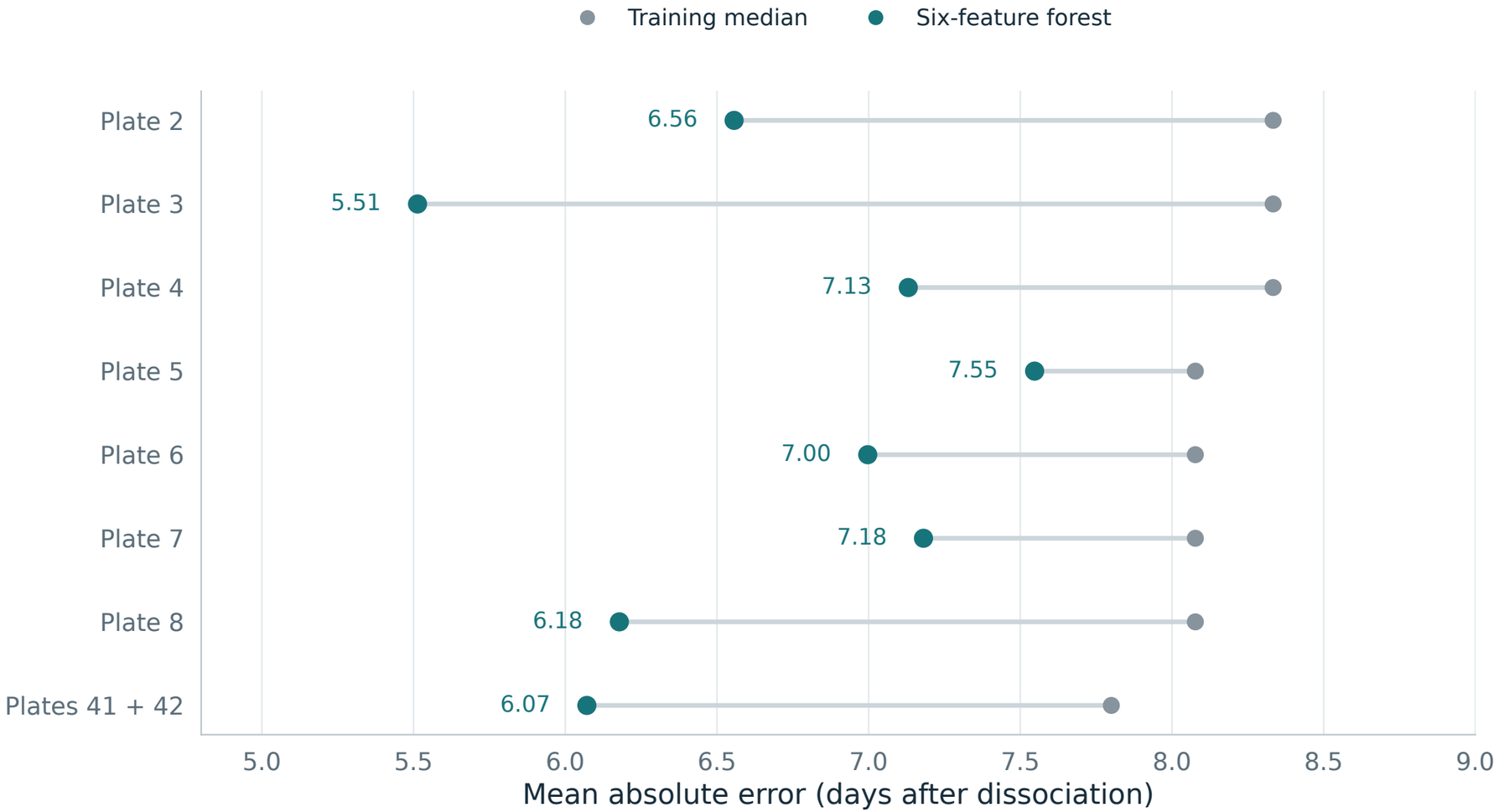
Change in MAE after removing one feature family from the full activity Ridge model.



Source: prespecified one-hour Ridge ablations, reference computation. Negative change means lower error on this inspected transfer set. This is not causal feature importance.

Age-benchmark performance varies across held-out groups

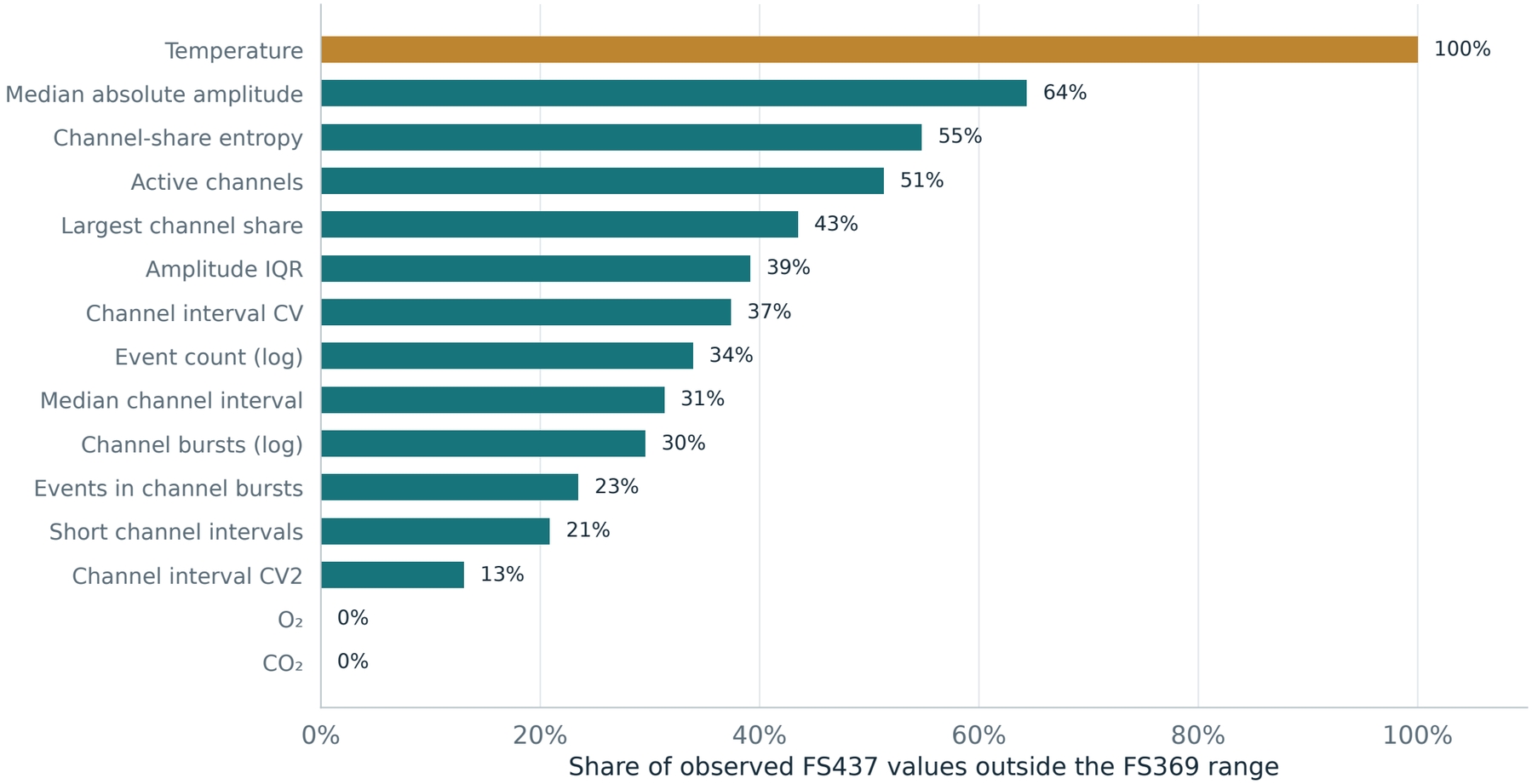
Each row is a full primary validation group. Every forest test-fold R^2 is positive in this benchmark.



Source: eight primary plate/batch folds. These are fold-level errors, not independent well-level confidence intervals. Plates 41 and 42 form one group.

Many FS437 feature values lie outside the FS369 range

Fraction of nonmissing test observations below the training minimum or above the training maximum.



Source: hourly feature-range audit. An out-of-range fraction describes observed dataset shift; it does not identify its cause or establish biological deterioration.